

VLSI Internship – Task 1 Report

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Internship:

VLSI (Very Large Scale Integration) Internship

Task 1 – Introduction to VLSI Design Flow and Basic Digital Logic Implementation

VLSI Design Flow

1. System Specification
2. Architecture Design
3. RTL Design
4. Functional Verification
5. Synthesis
6. Physical Design
7. Timing and Power Analysis
8. Fabrication
9. Testing and Validation

Truth Tables

AND Gate

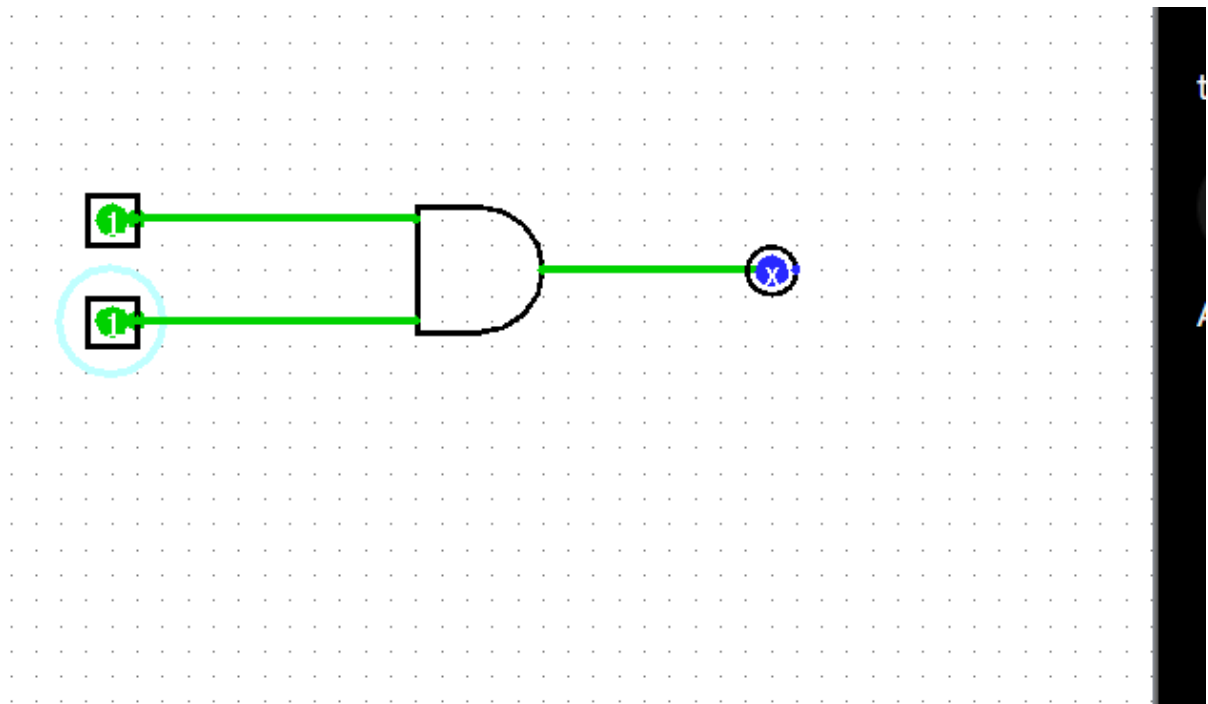
A	B	Output
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0	0	0
---	---	---

0	1	0
---	---	---

1	0	0
---	---	---

1	1	1
---	---	---



OR Gate

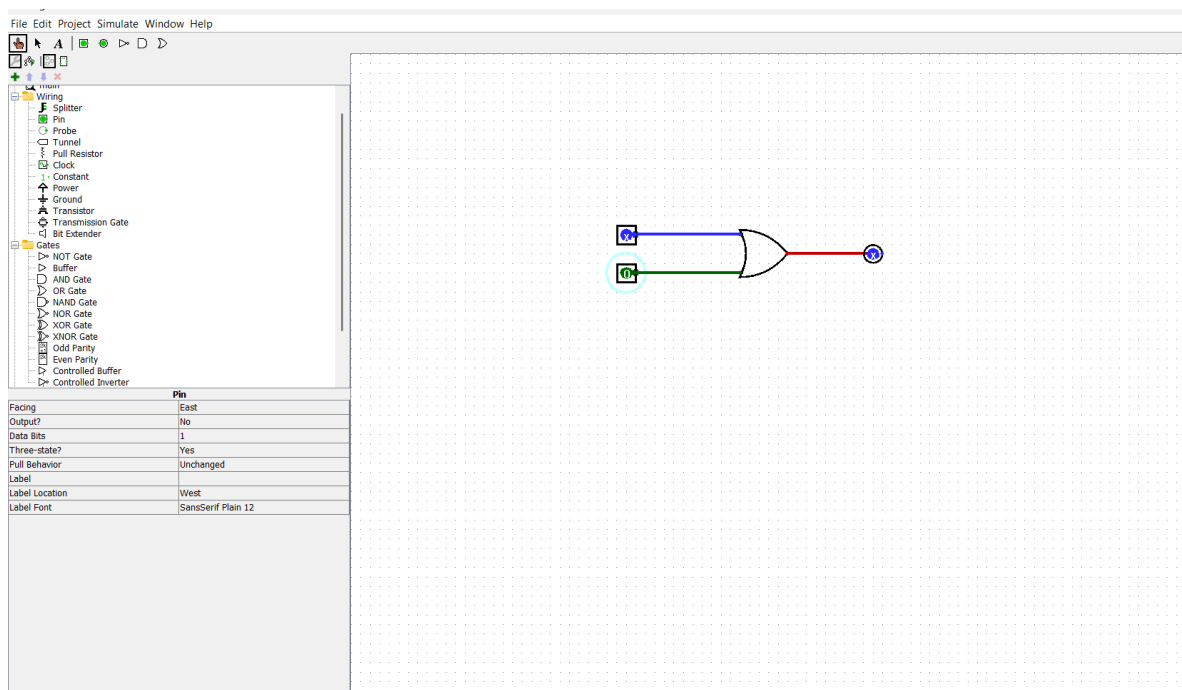
A B Output

0 0 0

0 1 1

1 0 1

1 1 1

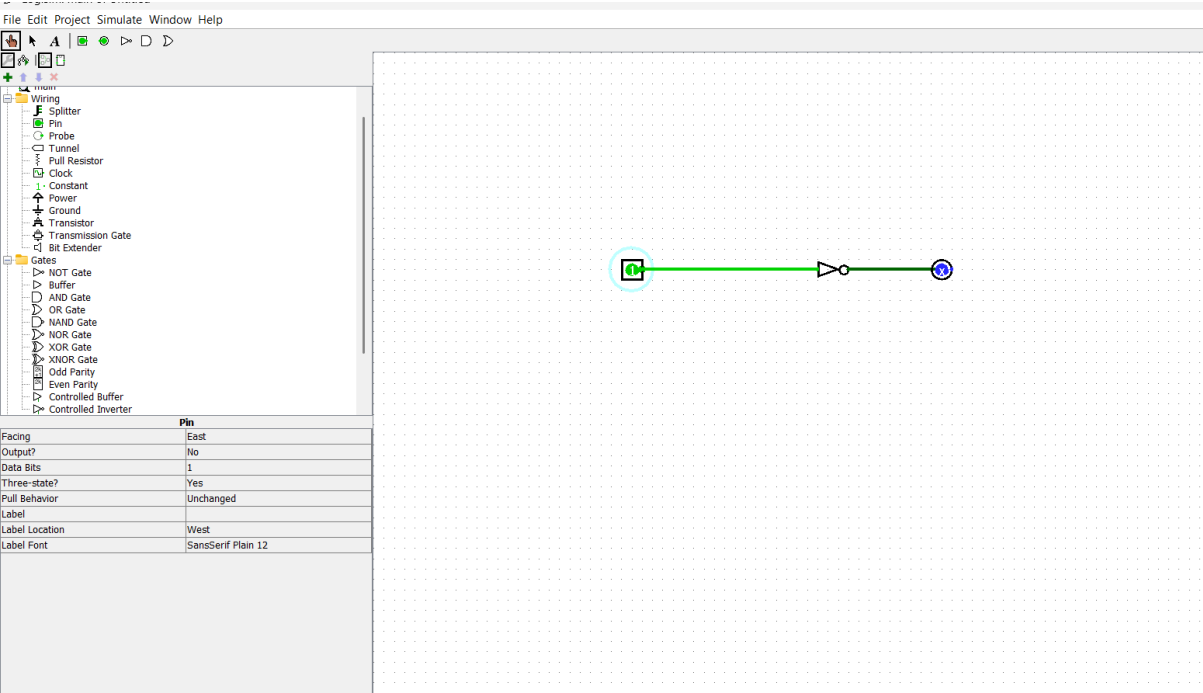


NOT Gate

A Output

0 1

1 0



NAND Gate

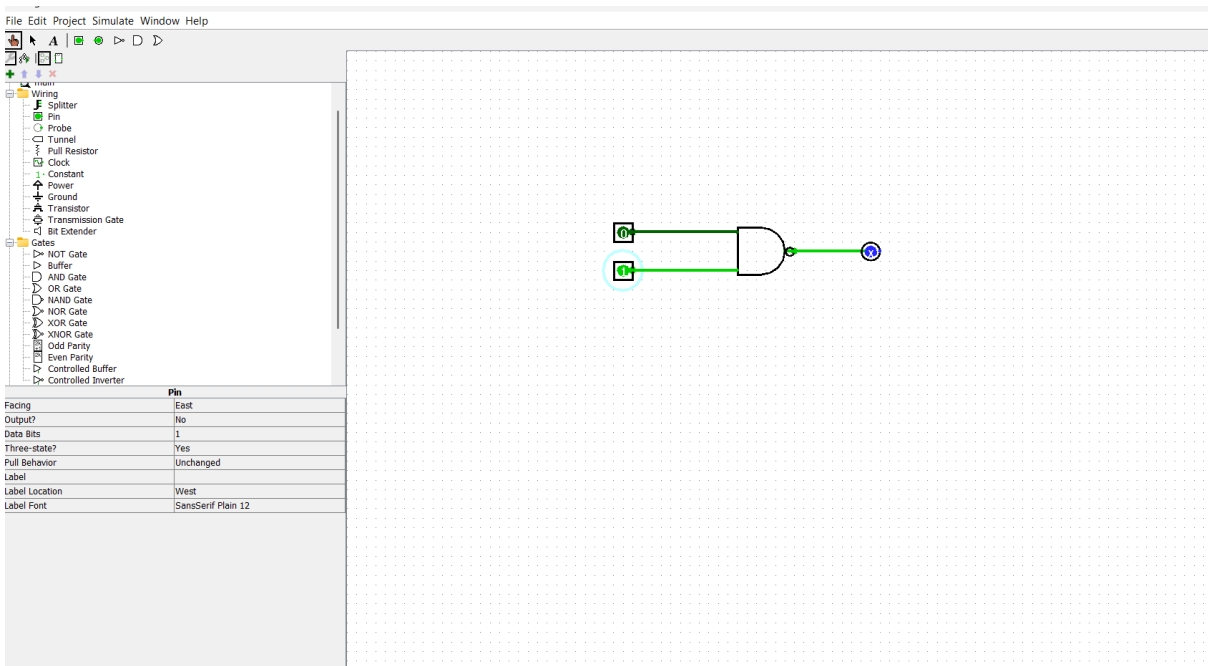
A B Output

0 0 1

0 1 1

1 0 1

1 1 0



NOR Gate

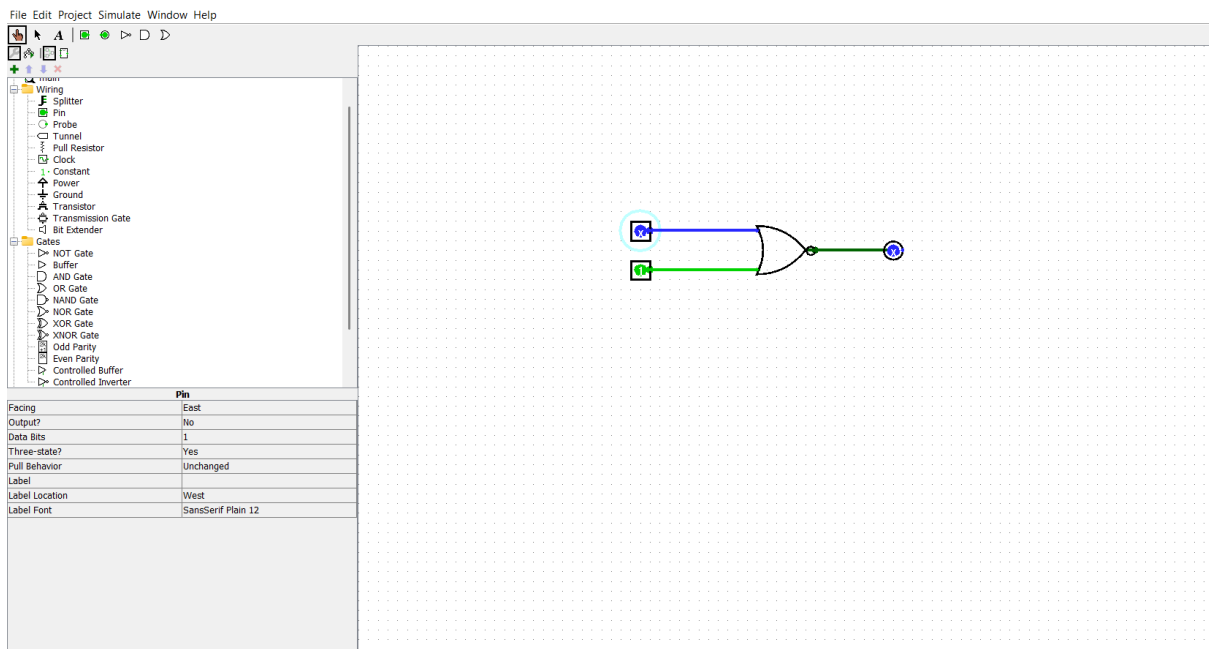
A B Output

0 0 1

0 1 0

1 0 0

1 1 0



XOR Gate

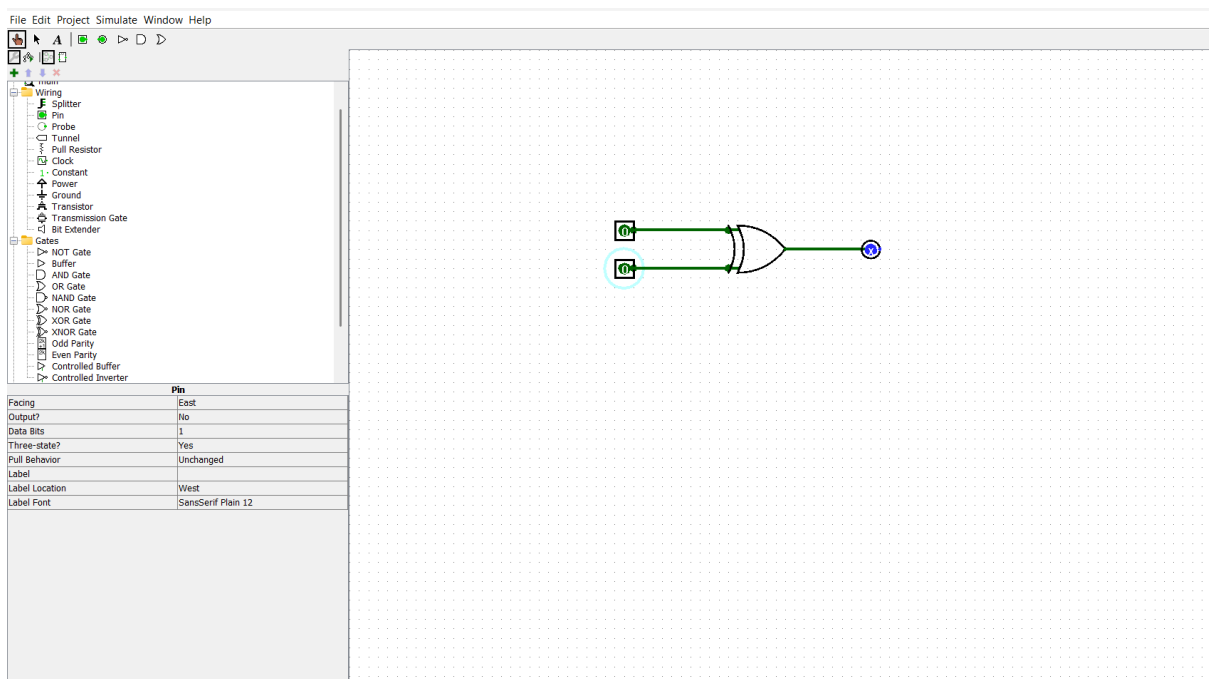
A B Output

0 0 0

0 1 1

1 0 1

1 1 0



Half Adder

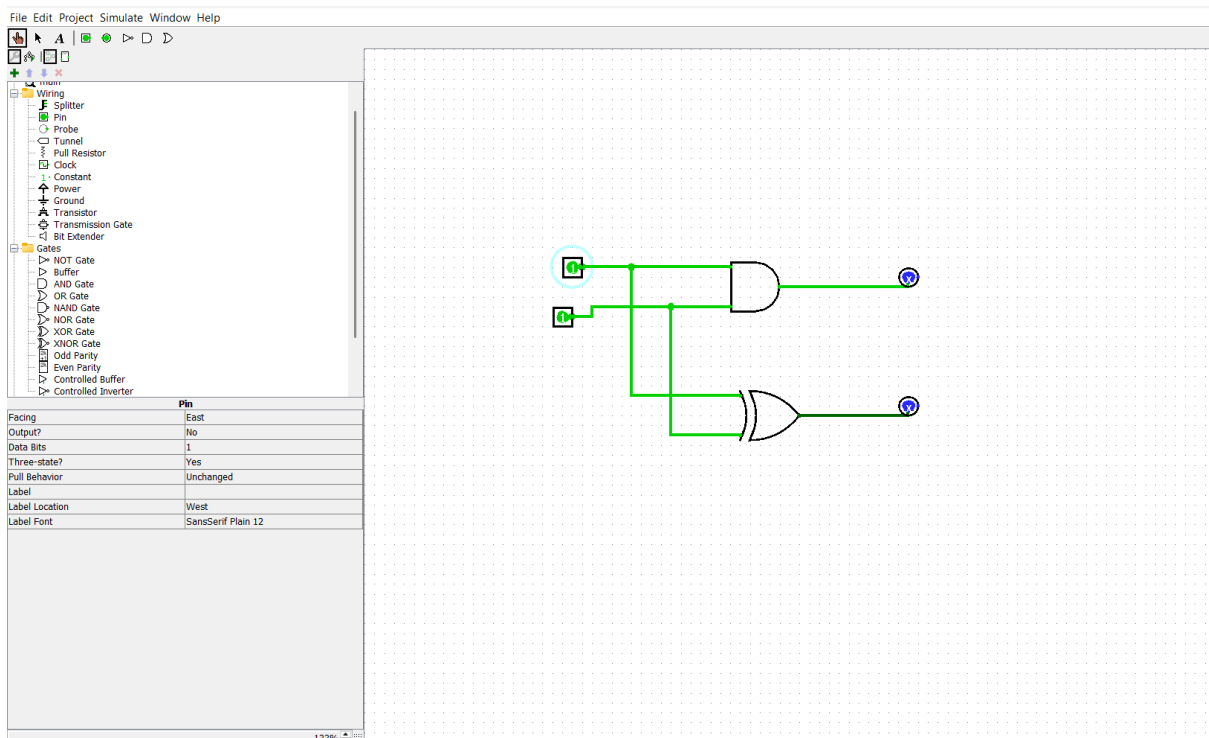
A B Sum Carry

0 0 0 0

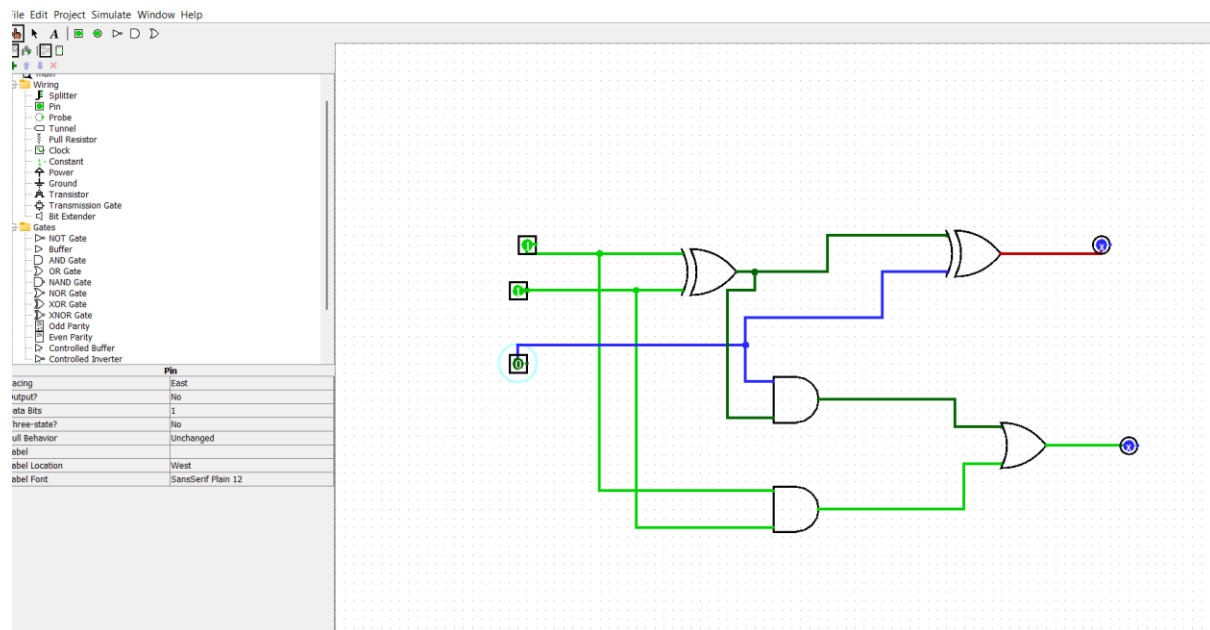
0 1 1 0

1 0 1 0

1 1 0 1



FULL ADDER



Observations

- All logic gates were implemented and simulated successfully using Logisim.
- Outputs matched the corresponding truth tables.
- The Half Adder circuit was implemented using XOR and AND gates.
- Sum output was obtained from the XOR gate.
- Carry output was obtained from the AND gate.
- Simulation results verified the correct operation of the circuits.